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Machine Learning Techniques

Unit-2 + PYQs.

Topics

1. Regression and its type? linear Vs logistic Regression. + 2022-23 [10 marks]
2. Bayesian learning + Baye's Thiarem + [2022-23, 2020-21]
3. Baye's optimal classifier ✓
4. Naive Bayes classifier + 2021-22 [10 marks]
5. E-M Algo - 2021-22 ✓
6. Concept learning + 2021-22 ✓
7. Support Vector Machine. + 2022-23, 2020-21 ✓
   - → Types
   - → Kernel function and its type ✓
   - → properties
   - → Issues.

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Road to 5K

* Regression - Unit-1

- It is a statistical model/method used to model the relationship b/w a dependent variable and one or more independent variables.
- The model can take various forms, such as linear regression, logistic regression, and polynomial regression, depending on the nature of the relationship b/w the variables.

* Linear Regression

↳ Same definition as Regression.

→ It makes predictions for continous/real or numeric values such as sales, age, product price, etc.

→ The goal is to find the line of best fit.

→ Simple linear equation = $y = mx + b$

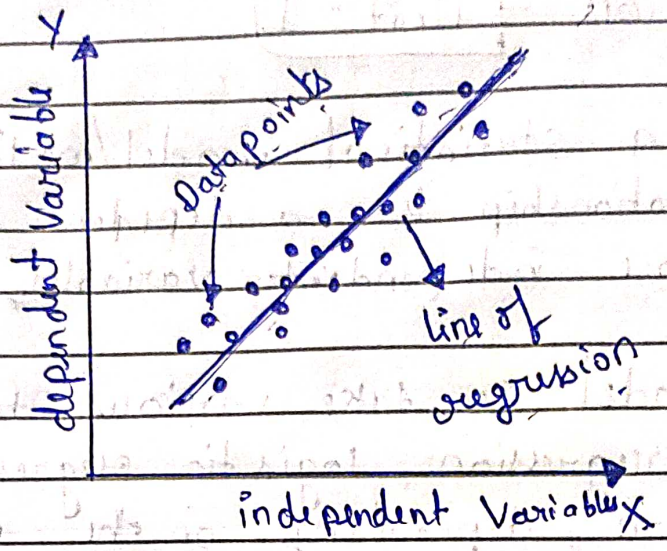
→ $Y \rightarrow$ dependent and x is independent, $m \rightarrow$ slope of line.

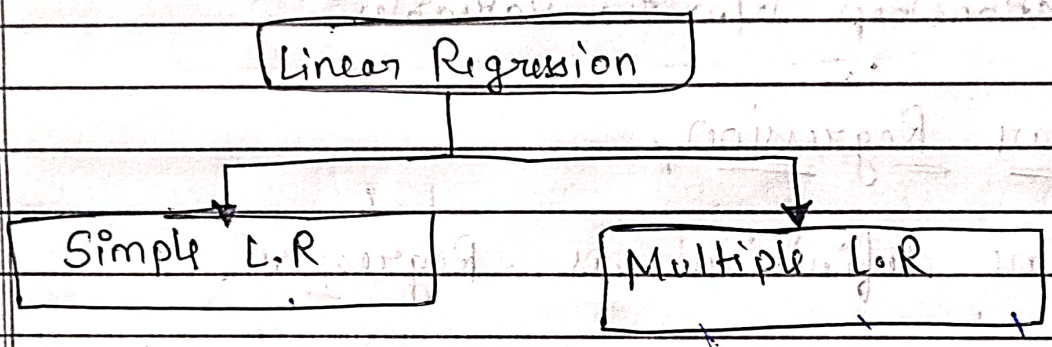
→ It can be used for both prediction and Inference (Assumption).

→ The linear regression model provides a sloped straight line representing the relationship b/w the variables.

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① Simple Linear Regression ⇒ If a single independent variable is used to predict the value of a numerical dependent variable.

eqv → $Y = b_0 + b_1 * x_i$

Y → dependent variable, x → independent
b1 → slope of the line.

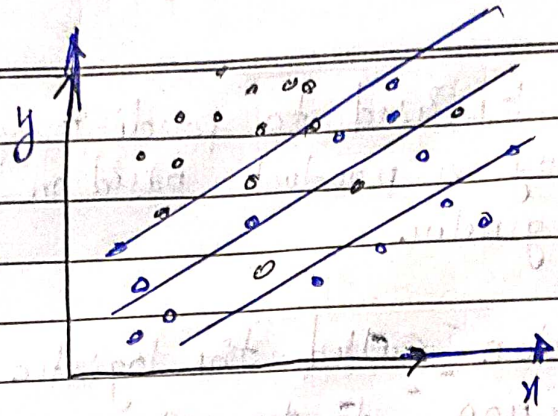
② Multiple linear Regression ⇒ If more than one independent variable is used to predict the value of numerical dependent variable.

eqv → $Y = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_n * x_n$

Y → dep, x → indep, b_1, b_2 → slope.

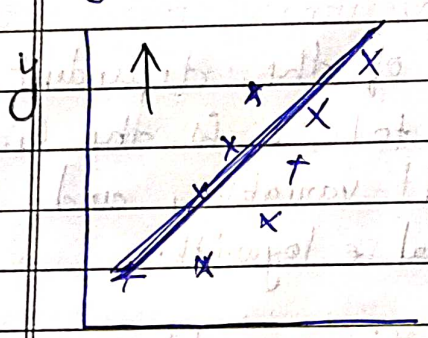
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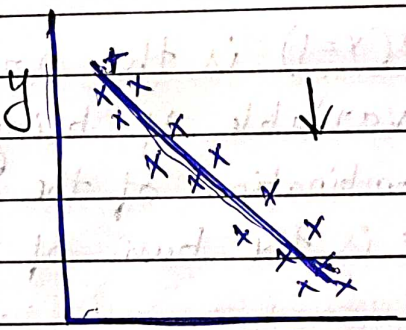


* Linear Regression line ⇒ It showing the r/p b/w the dependent and independent variables called a regression line.

→ Types of relationship:



Positive Correlation.



Negative Correlation.

⇒ Logistic Regression

→ Same as linear Regression but here the dependent variable is binary or categorical.
e.g ⇒ 0101010001
0110101110

→ It is used to predict the probality of an event occurring based on the values on the independent variables.

→ Two possible outcome → True or false. or 0 or 1.

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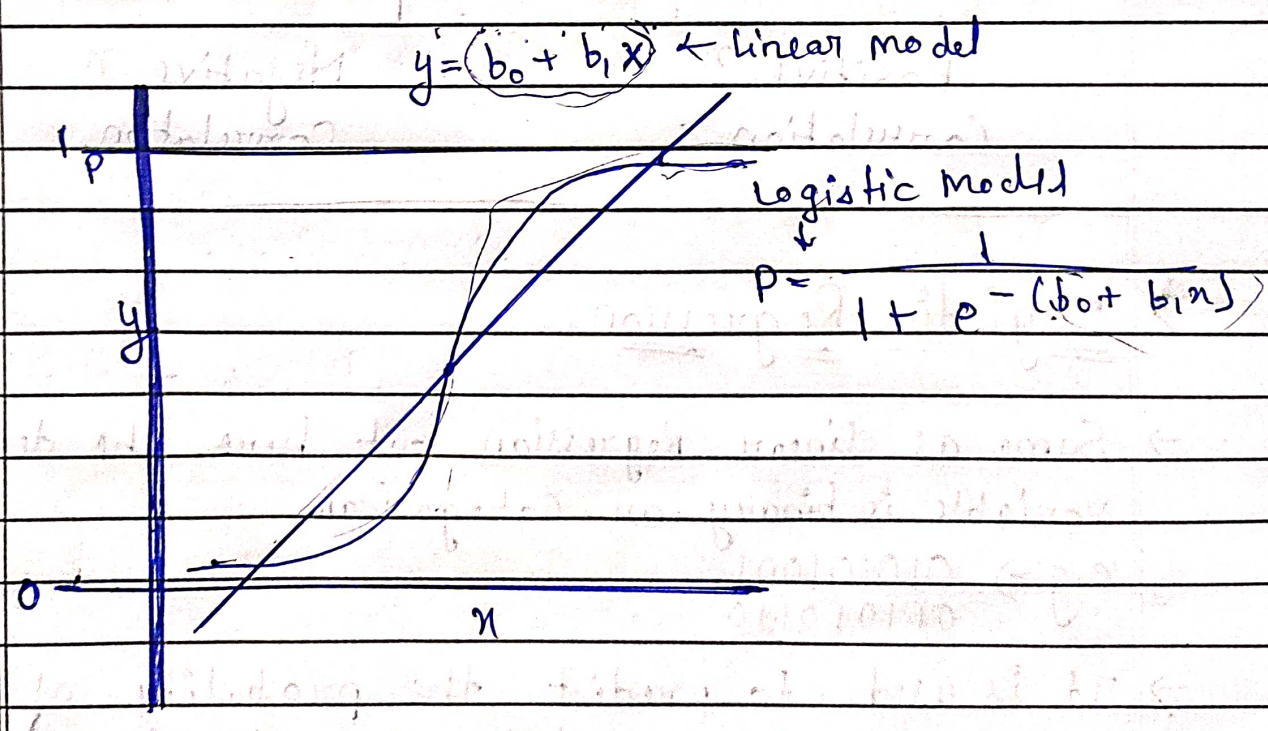
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→ for eg → it could be used to predict whether a customer will buy a product based on their age, income, and gender.

→ It uses a function called the logistic function or sigmoid function to transform the output of a linear equation into a value b/w 0 and 1

$$\text{equ} \Rightarrow P(Y=1) = 1/(1+e^{-z})$$

P(Y=1) is the probability of the dependent variable y being equal to 1, z is the linear combination of the independent variables, and e is the base of the natural logarithm



fields ⇒ healthcare, finance and marketing.

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* Bayesian Learning ✓

→ B.L is a fundamental statistical approach to the problem of pattern classification.
→ In B.L., a model is specified with parameters that describes the probability distribution of the data.
→ These parameters are initially set using prior knowledge or assumptions, and then updated based on the observed data using Baye's theorem.
→ The updated probabilities are used to make predictions about new data.

→ It useful where is data is limited or noisy

Application ⇒ spam filtering ✓
⇒ medical diagonosis ✓
⇒ NLP ✓
⇒ image recognition etc. ✓

* Baye's Theorem - AKTU - 2022-23, 2020-21 ✓

→ It is a mathematical concept that describes the relationships b/w conditional probabilities of two events.

$$P(A|B) = \frac{P(B|A)\ P(A)}{P(B)}$$

→ It is used in A.I and M.L. to build probabilistic models and make decisions based on uncertain data.

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Likelihood (PRIOR)

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

Posterior (The probability of A on Event B.)

(Marginalization) Probability of B,

* Concept learning - AKTU - 2021-22.

→ It refers to the process of building a probabilistic model that can recognize and generalize pattern in data and use that knowledge to classify new examples.

→ Bayesian Concept learning involves using Bayesian inference.

→ B.C.L involve

① Defining a PRIOR probability distribution.

② Collecting a set of training examples

③ Using the training examples and Baye's theorum.

④ Using the POSTERIOR distribution to make prediction.

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* Bayes optimal classifier

→ The Baye's optimal classifier is a probabilistic model that predicts the most probable outcome for a new instance using the Bayes Theorem.

* Naive Bayes Classifier AKTU-2021-22

→ A N.B.C. is a probabilistic machine learning model that's used for classification task. The base of the classifier is Bayes Theorem.

→ N.B. Cs are a collection of classification algos.

→ It is not a single algorithm but a family of algorithms where all of them share a common principle.

- Assumption = The fundamental Naive Bayes assumption is that each features makes an Independent, Equal contribution to the outcome.

- Note: The assumptions made by Naive Bayes are not generally correct in real-world situations.

→ Write Bayes theorem formula and explain it.

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AKTU - 2021-22 [10 marks]

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E-M Algorithm

→ The E-M (Expectation - Maximization) algorithm is an iterative optimization method used to estimate the parameters of probabilitic models when there are missing or incomplete data.

→ It is a technique to find maximum likelihood estimation when the latent variables are present.

→ It is also refferred to as the latent variable model.

→ The EM algo is combination of various unsupervised machine learning algo.

Steps.

① Initialization : start by initializing the parameters of the model with some initial values.

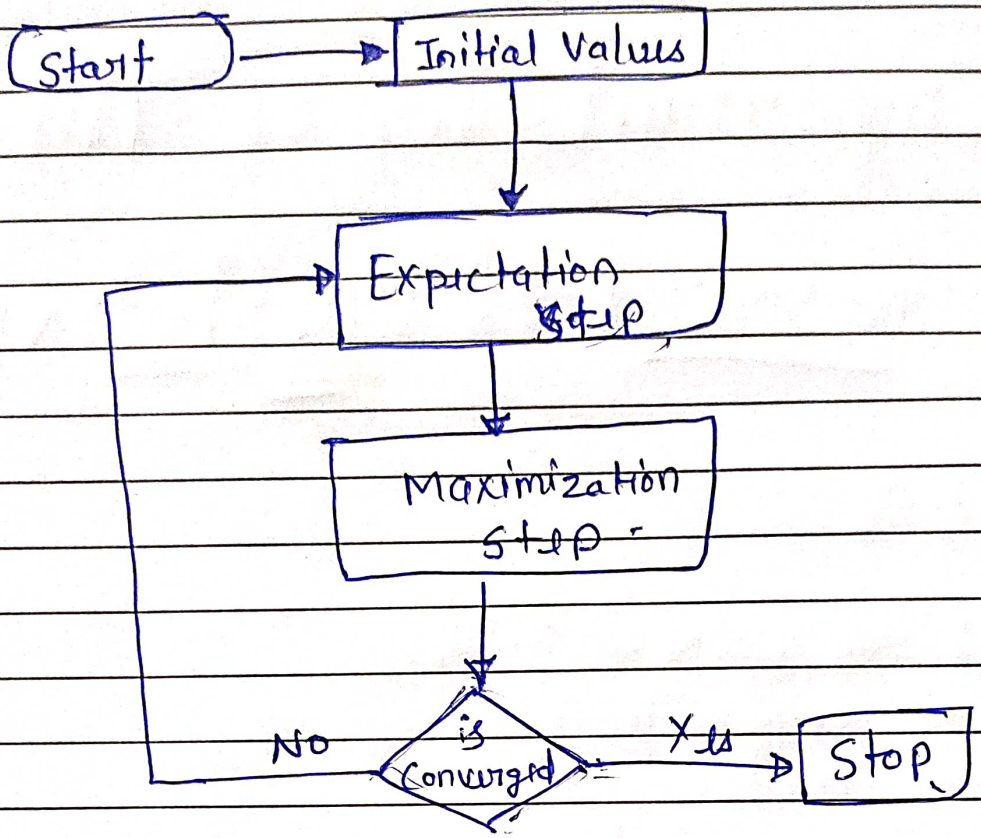
② Expectation step(E): Calculate probability.

③ Maximization step (M): Updating parameter.

④ Iteration : Repeat E-step and M-step until the convergence of the values occurs.

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Start
Initial Values
Expectation Step
Maximization Step
is Converged
No
Yes
Stop

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* Support Vector Machine

→ SVM is a type of supervised machine learning algorithm that can be used for classification or regression tasks.

→ SVM tries to find the best possible boundry between the different classes in the data by finding a hyper plane that maximizes the margin b/w two or more classes.

→ It is often used in applications such as image recognition, text classification etc.

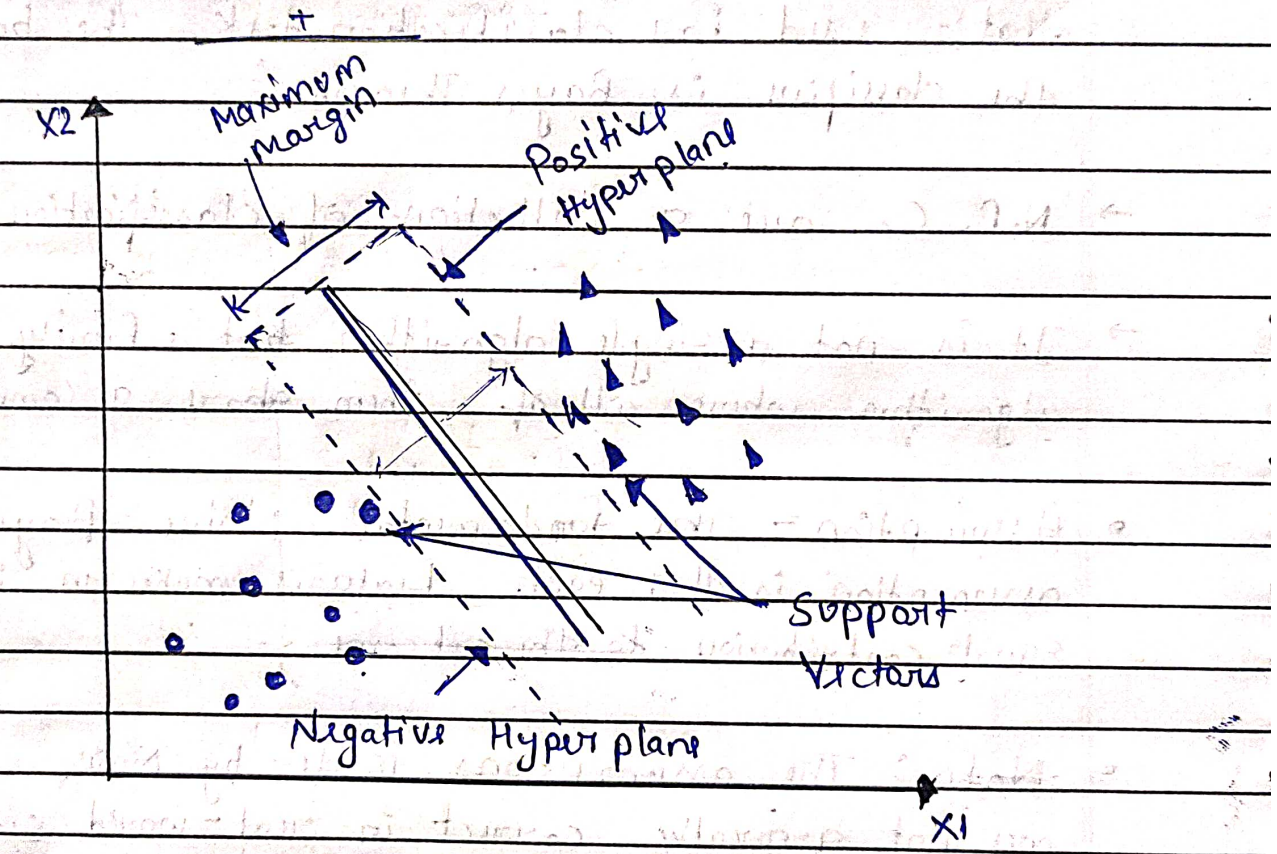
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* EM algorithm and its steps → Read ~~from~~ ~~Notes~~ AKTU - 2022-23, 2020-21 (2 times 10 marks).

* Support Vector Machine

→ Definition from Unit-1

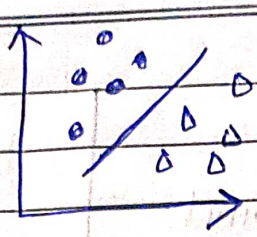


* Types of SVM

1. Linear SVM

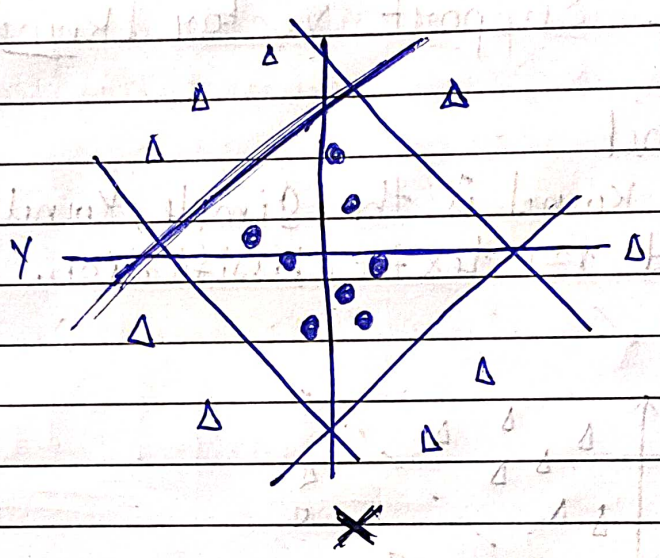
→ It is used for linearly separable data, what means dataset divided into two classes by using single straight line.

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② Non-Linear SVM

→ It is used for non-linearly separated data, which means if a dataset cannot be classified by using straight line.



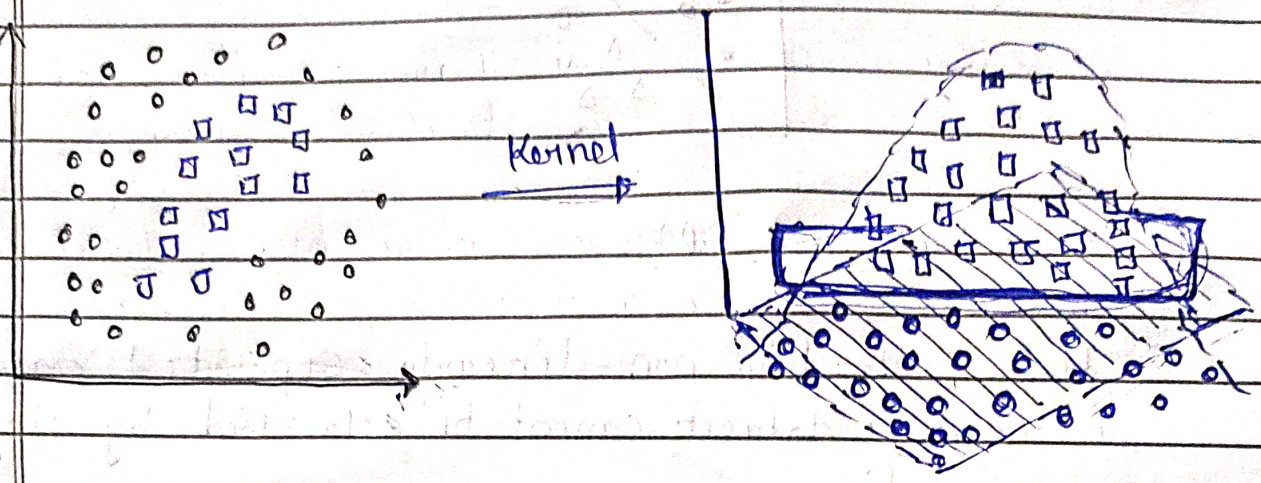
* Kernel ~ AKTU - 10 marks

→ It is function that is used to transform the original feature space into a higher-dimensional space.

→ The Kernel function enables ~~SVC~~ SVM to effectively handle non-linear classification problems. This allows SVM to find a non-linear decision boundary in the transformed feature space, even through the original data may not be linearly separable.

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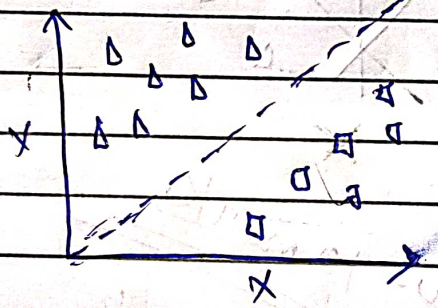
* Types of Support Vector Kernel ✓

① Linear Kernel

→ The linear kernel is the simple kernel function.

→ mostly used for text-classification.

→ straight line ✓



② Polynomial Kernel

→ It introduces new dimensions by computing all possible polynomials up to a specified degree of the original features.

→ It captures non-linear relationships b/w the data points.

Diagram

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(3) Gaussian Kernel.

→ also known as Radial Basis function (RBF) Kernel.
→ widely used and is effective for handling non-linearly separable data.
→ It maps the data into an infinite-dimensional space where each data point is represented as a Gaussian function.

Properties of SVM.

1. Maximal Margin. → distance b/w hyperplane.
2. Non-linearity with Kernels.
3. Support Kernel. → Dataset.
4. SVB also work on small number of support vectors.
5. Versatility → (use for both Regression and classifica
6. Control of Complexity → (user can control it's Complexity)

Issues in SVM

1. Scalability with large Dataset ⇒ Required more cost and memory-intensive.

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② Lack of Probabilistic Interpretation : (does not provide probabilities outputs)

③ Imbalanced Data : where the number of samples in different classes is significantly unequal, SVMs may be biased towards the majority class.

④ Interpretability : SVMs tend to provide good predictive performance, but they may not offer direct interpretability of the learned model.

Unit-2 - Completed.

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